

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-007 - Nahorkatiya-007

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000074
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2002-04-29 / 2002-06-22
Measured / TVD	2062 m / 2034 m
Trajectory	Vertical; inc 5.1 deg; az 259.1 deg
Formation Interval	1220-1840 m
Lithology	Claystone-sandstone alternation
Pore Pressure Reference	2388 psi
Fracture Gradient	14.55 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Low (34.7/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	515	Surface
9 5/8 in casing	1195	Intermediate
7 in liner/casing	1814	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2002-05-21 956 m	Pressure Spike	Critical	Circulating pressure climbed without equivalent torque change. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2002-05-11 569 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Condition hole and confirm pressure normalization. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2002-05-30 1264 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2002-05-05 338 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.
2002-05-05 357 m	Torque Spike	Medium	High torque developed while WOB remained elevated. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Use offset torque trend to set adaptive operating envelope.
2002-05-14 706 m	Cementing Issue	Medium	Cement returns were below planned percentage. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Offset loss history should inform cementing hydraulics.
2002-05-25 1087 m	Cementing Issue	Medium	Cement returns were below planned percentage. Mitigation: Review loss history before cementing equivalent interval. Lesson: Offset loss history should inform cementing hydraulics.

Completion Summary

The synthetic well reached 2062 m MD in the Girujan interval. The NWIS knowledge record links 8 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.